

Pediatric lead testing disparities in New York State

A retrospective analysis using biostatistic principles in R

Cameron Lincoln - cal386

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1 Introduction

1.1 Scientific Plausibility, Context, and Research Question

Throughout several US regions and states, public health campaigns ensure that children are routinely screened for blood lead levels. Due to a variety of social, environmental, and economic factors, children of historically marginalized communities are at higher risk of lead exposure, which can lead to adverse developmental health impacts.¹ Several states have internal guidelines and requirements on blood lead testing for children. In the state of New York, all children are expected to undergo regular checkups to screen for elevated blood lead testing.² These guidelines are subject to political or administrative instability, and may change frequently. For New York, medical providers had different testing requirements as of 2019, and property rental registry laws were updated in 2025.² On the national level, there have been efforts to improve drinking water via the 2021 Infrastructure Investment and Jobs Act.³ However, these waterline projects are managed by individual states to determine what projects are prioritized.³ As such, there is no suggested guideline to systematically address disparities in lead exposure in water ways, leading to inconsistencies across state lines. For New York State, municipalities with high populations of people of color and lower incomes are likely to be prioritized by the lead service line replacement program, as these individuals may be more at risk for blood lead levels.³

This analysis hopes to establish a link between geography and blood lead levels in children across New York State. It will explore if changes are observed in **reported positive cases of elevated pediatric blood lead level (varA)** over **time in years (varB)** across **different counties of New York State, excluding New York City (varC)**.

2 Methods

This analysis will hinge on data gathered by the New York State Department of Health (NYSDOH). Per state regulation, doctors are required to do blood level testing for all children at specific time intervals. This data includes all children tested for lead from 2000 to March 6, 2024. Data was merged with Federal Information Processing Standards (FIPS) codes by ZIP code and county for generating visuals.

2.1 Important considerations about data quality

This data set does not include tests from NYC. Furthermore, this data set only captures individuals who reported to clinics in New York State; should an individual not have health insurance, seek care outside of New York, or forgo routine checkups altogether, this data set will not capture these children. Therefore, it should be understood that the data presented by NYSDOH is likely an underestimate of the true count of pediatric lead cases.

This data set presents nullification criterium to protect patient anonymity in cases where positive result/number of tests can identify individuals in low population areas. Tests are pooled by year, and are reported by ZIP code, geographic center, and county level. No demographics information was gathered. This data identifies the patient by testing ZIP code/county, not the patient's home location. This provides a level of uncertainty of the data in cases where patients seek out testing outside of their regular ZIP code, county, and/or state residence.

2.2 Accessing the data

To see the more information about the data set from the publisher, please visit the website provided: [at the NYDOH website](#). This data can be downloaded at [the following GitHub repository folder](#).

The following details the process of identifying variables for this analysis:

```
# Set up

setwd("/Users/clincoln/Desktop/vtpeh-6270")
library(ggplot2)
library(dplyr)
library(tidy)
library(knitr)
library(kableExtra)
library(gtsummary)
data_lead <-
  read.csv(
    "Childhood_Blood_Lead_Testing_and_Elevated_Incidence_by_Zip_Code__Beginning_2000_20260205.csv")

# Data previewing
head(data_lead)
colnames(data_lead)
summary(data_lead)
```

```

# Data cleaning
data_lead$County.Code <- as.character(data_lead$County.Code)
data_lead$Zip <- as.character(data_lead$Zip)
data_lead$Tests <- as.numeric(data_lead$Tests)
data_lead$Less.than.5.mcg.dL <- as.numeric(data_lead$Less.than.5.mcg.dL)

# So from the data set, I want to use: county, county code, year, ZIP code,
# number of tests performed, percentage, and rate per 1,000
# Before I go in and change the names, I'm going back to the data source to
# double check that my re-names are in alignment with the codebook.

# Renaming all my columns to be consistent. NOTE: Percent needs to be
# "New_Case_Percentage," since these ONLY count newly diagnosed children who
# had a positive blood Pb level.

data_lead_tidy <- data_lead
data_lead_tidy <- data_lead %>%
  select(-"Less.than.5.mcg.dL", -"X5.10.mcg.dL",
        -"X10...15.mcg.dL", -"X15...mcg.dL")

data_lead_tidy <- data_lead_tidy %>%
  rename("County_Code" = "County.Code",
        "Total_Elevated_Blood_Levels" = "Total.Elevated.Blood.Levels.",
        "New_Case_Percentage" = "Percent",
        "Rate_per_1000" = "Rate.per.1.000",
        "ZIP_Location" = "Zip.Code.Location",
        "County_Location" = "County.Location")

# I'm going to exclude the 0-data
table(data_lead_tidy$Rate_per_1000)
non0rate_data <- data_lead_tidy[
  data_lead_tidy$Rate_per_1000 > 0,]
table(non0rate_data$Rate_per_1000)

# This should give us a much cleaner data set.
head(non0rate_data)
summary(non0rate_data)

```

The following code will generate a table of the variables used in this data set:

```

data_lead_vars <- data.frame(
  Variable = c("County",
              "County_Code",
              "Year",
              "Zip",
              "Tests",
              "Total_Elevated_Blood_Levels",
              "New_Case_Percentage",
              "Rate_per_1000",
              "ZIP_Location",
              "County_Location"),
  Type = c("Categorical",
           "Categorical",

```

Table 1: Table with all the variables by their class and type

Variable	Type	Class
County	Categorical	Character
County_Code	Categorical	Character
Year	Discrete	Integer
Zip	Categorical	Character
Tests	Discrete	Numeric
Total_Elevated_Blood_Levels	Continuous	Integer
New_Case_Percentage	Continuous	Numeric
Rate_per_1000	Continuous	Integer
ZIP_Location	Categorical	Character
County_Location	Categorical	Character

```

        "Discrete",
        "Categorical",
        "Discrete",
        "Continuous",
        "Continuous",
        "Continuous",
        "Categorical",
        "Categorical"),
Class = c("Character",
          "Character",
          "Integer",
          "Character",
          "Numeric",
          "Integer",
          "Numeric",
          "Integer",
          "Integer",
          "Character",
          "Character")
)

kbl(data_lead_vars[1:10,], booktabs = T,
     caption = "Table with all the variables by their class and type") %>%
  kable_styling(latex_options = c("striped", "scale_down"))

```

Here, there are both categorical and continuous variables of interest. To answer the questions set out in the introduction, the following variables were selected:

- **County**, a categorical variable with 57 unique counties within New York State, excluding New York City
- **Year**, a discrete variable from any test/value that occurred within January 1 to December 31 of each time interval.
- **Total_Elevated_Blood_Levels**, a discrete variable with the number of individuals who tested positive for elevated blood levels.
- **Tests**, a discrete variable with the number of individuals who tested for lead (per county)

- **Rate_per_1000**, a continuous variable defined per 1,000 cases (within each year/county/etc.). This was calculated by dividing **Total_Elevated_Blood_Levels** by **Test**. This was selected in order to compare across different counties, these rates may bias towards larger counties with more capacity to test.

3 Results

The following details the data analysis from the full data comparing Year, County, and Rate_per_1000.

3.1 Graphics

The following code will generate scatter plots with jitter to display the spread of data by county, year, and number of cases. This data will also generate a table of values displaying the median rate, IQR, and number of within group observations.

```
# Summary statistics by county
library(forcats)
county_summary <- non0rate_data |>
  group_by(County) |>
  summarise(
    median_rate = median(Rate_per_1000, na.rm = TRUE),
    q25          = quantile(Rate_per_1000, 0.25, na.rm = TRUE),
    q75          = quantile(Rate_per_1000, 0.75, na.rm = TRUE),
    n_obs        = n(),
    .groups      = "drop"
  ) |>
  arrange(median_rate) |>
  mutate(County = fct_inorder(County))

county_summary |>
  mutate(
    median_rate = round(median_rate, 1),
    q25          = round(q25, 1),
    q75          = round(q75, 1),
    IQR          = paste0(q25, "-", q75) # collapse Q1/Q3 into one column
  ) |>
  select(County, median_rate, IQR, n_obs) |>
  kable(
    format = "latex",
    booktabs = TRUE,
    longtable = TRUE, # allows table to span multiple pages
    col.names = c("County", "Median Rate (per 1,000)",
                  "IQR (Q1-Q3)", "Observations (n)"),
    caption = "Summary of blood lead rates per 1,000 by county, ordered by median rate",
    align = c("l", "c", "c", "c")
  ) |>
  kable_styling(
    latex_options = c("striped", "hold_position", "repeat_header"),
    font_size      = 9 # smaller font keeps 50+ rows manageable
  )
```

Table 2: Summary of blood lead rates per 1,000 by county, ordered by median rate

County	Median Rate (per 1,000)	IQR (Q1-Q3)	Observations (n)
Rockland	7.0	5-9.5	19
Nassau	11.0	9-93.5	27
Suffolk	11.0	8-20	5
Westchester	15.0	11-20	71
Jefferson	15.5	14.5-18.2	12
Cayuga	23.0	21.8-24.8	12
Dutchess	25.0	16-33	17
Cortland	26.0	23.8-28	4
Oswego	27.0	24-28	5
Chautauqua	29.0	22.5-36.5	23
Ontario	30.5	29.5-33.8	4
Rensselaer	30.5	24.2-44	30
Chemung	33.0	28-45.5	8
Cattaraugus	34.0	33-36.5	6
Niagara	34.0	24-47	37
Orange	34.0	20.5-52.5	43
Broome	36.0	26.8-43	32
Monroe	36.0	26-48.2	84
Ulster	36.5	29-47.8	20
Genesee	37.0	35-37.5	3
Schenectady	37.0	30-50	25
Columbia	41.5	34-52.2	12
Herkimer	47.0	42.5-67.5	3
St. Lawrence	47.5	46.2-48.8	2
Montgomery	49.5	37.2-76	20
Albany	53.0	40-75.5	71
Onondaga	53.0	41.5-70	115
Steuben	54.0	39-57	9
Madison	56.0	56-56	1
Fulton	57.0	50-78	25
Oneida	61.0	38.5-90	63
Erie	63.0	44-92	189
Yates	71.0	71-71	1
Orleans	74.5	67.8-81.2	2
Warren	75.0	69-110	3
Sullivan	90.0	90-90	1
Otsego	194.0	194-194	1
NA	NA	NA-NA	6752

```
# scatter plot per county
ggplot(county_summary,
  aes(x = median_rate, y = County)) +
  geom_jitter(
    data = non0rate_data |>
      mutate(County = factor(County, levels = levels(county_summary$County))),
    aes(x = Rate_per_1000, y = County),
    width = 0, height = 0.2,
    colour = "#378ADD", alpha = 0.6, size = 0.4,
    inherit.aes = FALSE
```

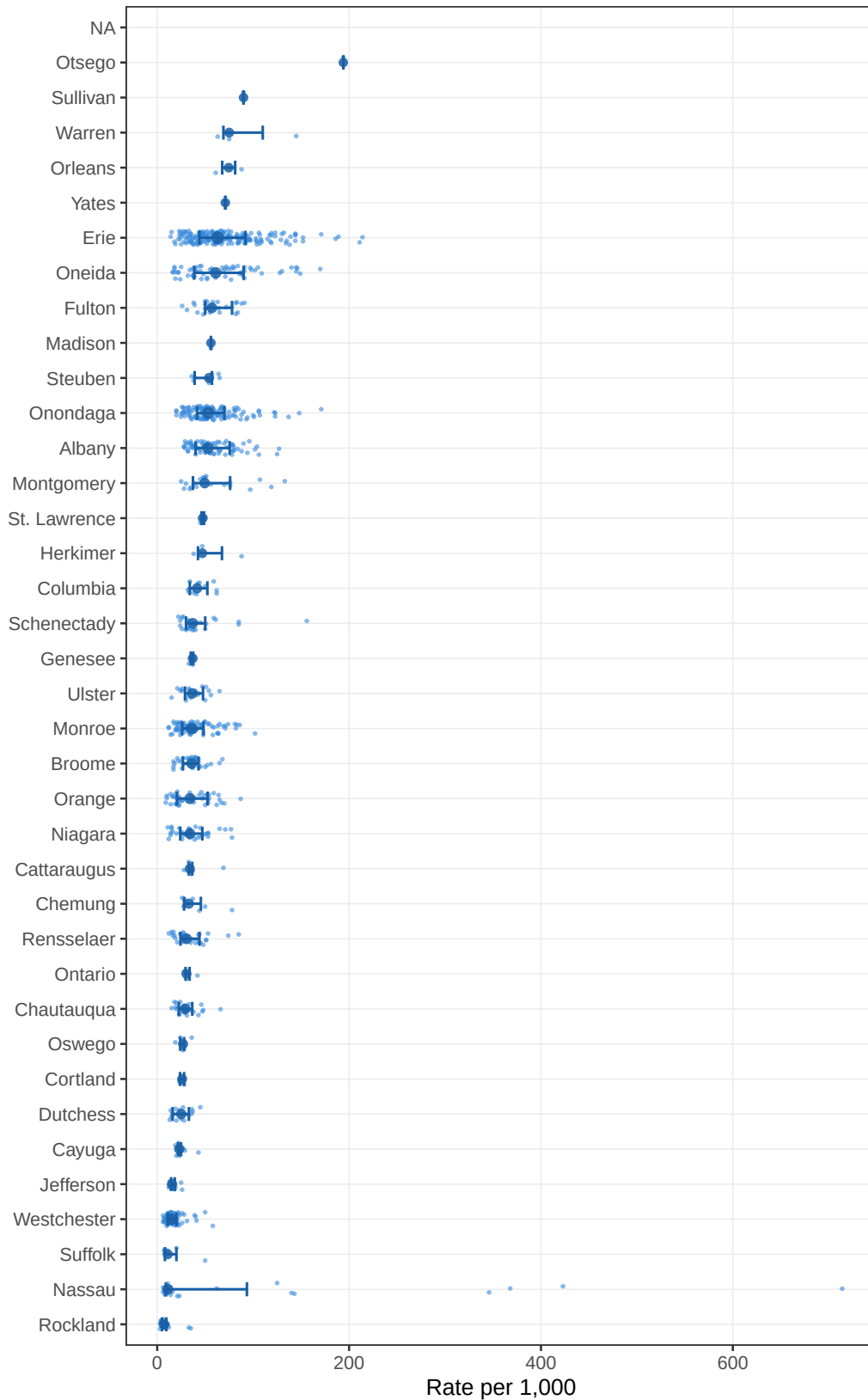
```

) +
geom_errorbarh(
  aes(xmin = q25, xmax = q75),
  height = 0.4, colour = "#185FA5", linewidth = 0.6
) +
geom_point(
  aes(size = n_obs),
  colour = "#185FA5", alpha = 0.85
) +
scale_size_continuous(
  name = "Observations (n)",
  range = c(1.5, 6),
  labels = scales::comma,
  guide = "none"
) +
labs(
  title = "Median blood lead rate per 1,000 by county",
  subtitle = "Bars show IQR (Q1-Q3); point size reflects n; dots are individual ZIP-year observations",
  x = "Rate per 1,000",
  y = NULL
) +
theme_bw(base_size = 11) +
theme(
  panel.grid.major.y = element_line(linewidth = 0.3),
  panel.grid.major.x = element_line(linewidth = 0.3),
  panel.grid.minor = element_blank(),
  legend.position = "bottom"
)

```

Median blood lead rate per 1,000 by county

Bars show IQR (Q1–Q3); point size reflects n; dots are individual ZIP–year c



4 Statistical analysis and discussion

The following details the process of analyzing the data by checking for assumptions. Data was filtered to exclude observations with zero cases reported, as this could reduce ambiguity in the analysis. Counties/ZIP codes with NA or “zero” values could be attributed to a lack of testing being performed or no lead being detected.

Normality of the rate per 1,000 cases (`Rate_per_1000`) was assessed using a Kolmogorov-Smirnov test, since the number of observations ($n = 7,757$) is over the Shapiro-Wilks limit of 5,000 observations. A significantly significant result ($D = 0.1409$, $p < 0.001$) indicates that normality can not be assumed. Therefore, non-parametric tests were selected moving forward.

To analyze differences across counties, a Kruskal-Wallis test was selected since it aligned with the assumptions made (non-parametric, independence, spread/distribution is similar between groups). Dunn’s post-hoc comparisons were used to compare across different counties, using the BH correction to control for larger county sizes. The Kruskal-Wallis test indicated a statistically significant difference in pediatric lead exposure cases per 1,000 across different counties ($H(36) = 432.519$, $p < 0.01$). This suggests that geographic location could be associated with the burden of lead exposure within New York State (excluding New York City).

After doing the pairwise post-hoc test, several significant pairwise county differences were noted, with all pairwise values reporting a $p < 0.05$. However, effect size estimation was relatively small ($\epsilon^2 = 0.056$), suggesting that while differences are statistically significant, differences by county only account for about 5% of the differences; there may be other confounding variables or better predictors (e.g., sample size, number of tests, etc.) that might better correlate with cases of pediatric lead levels. Because of this, the statistical utility of this analysis is heavily limited, and only should be used to draw qualitative conclusions.

```
library(dunn.test)

# Checking for normality
# because n = 7,757 Shapiro-Wilk's limit of 5,000, so we use the
# Kolmogorov-Smirnov test instead.

ks_result <- ks.test(
  non0rate_data$Rate_per_1000,
  "pnorm",
  mean = mean(non0rate_data$Rate_per_1000, na.rm = TRUE),
  sd    = sd(non0rate_data$Rate_per_1000, na.rm = TRUE)
)

cat("Kolmogorov-Smirnov normality test\n",
    "D =", round(ks_result$statistic, 4),
    "\np-value =", round(ks_result$p.value, 4), "\n\n")
```

Kolmogorov-Smirnov normality test D = 0.1409 p-value = 0

```
# Kruskal-Wallis test

kw_result <- kruskal.test(Rate_per_1000 ~ County,
                          data = non0rate_data)

cat("Kruskal-Wallis test\n",
    "H =", round(kw_result$statistic, 3),
    "\ndf =", kw_result$parameter,
    "\np-value =", round(kw_result$p.value, 4), "\n\n")
```

Kruskal-Wallis test $H = 432.519$ $df = 36$ $p\text{-value} = 0$

```
# Dunn's post-hoc test
# Performs pairwise comparisons between counties.

if (kw_result$p.value < 0.05) {

  dunn_result <- dunn.test(
    x      = non0rate_data$Rate_per_1000,
    g      = non0rate_data$County,
    method = "bh",
    alph   = 0.05,
    label  = TRUE
  )

  # Put into a tibble for making it easy for me to read
  dunn_tidy <- tibble(
    comparison = dunn_result$comparisons,
    Z          = round(dunn_result$Z,      3),
    p_adjusted = round(dunn_result$P.adjusted, 4),
    significant = ifelse(dunn_result$P.adjusted < 0.05, "Yes", "No")
  ) |>
  arrange(p_adjusted) |>
  filter(significant == "Yes")

  # kable
  dunn_tidy |>
  kable(
    format      = "latex",
    booktabs    = TRUE,
    longtable    = TRUE,
    col.names    = c("Comparison", "Z statistic",
                     "Adjusted p-value", "Significant"),
    caption     = "Dunn post-hoc test: significant county pairs (BH-corrected)"
  ) |>
  kable_styling(
    latex_options = c("striped", "hold_position", "repeat_header"),
    font_size     = 9
  )

} else {
  cat("KW test was not significant (p >= 0.05).\n",
      "Post-hoc testing not warranted.\n")
}
```

Table 3: Dunn post-hoc test: significant county pairs (BH-corrected)

Comparison	Z statistic	Adjusted p-value	Significant
Albany - Cayuga	4.535	0.0000	Yes
Albany - Chautauqua	4.559	0.0000	Yes
Albany - Dutchess	5.091	0.0000	Yes
Broome - Erie	-5.268	0.0000	Yes
Cayuga - Erie	-5.340	0.0000	Yes

Table 3: Dunn post-hoc test: significant county pairs (BH-corrected)
(continued)

Comparison	Z statistic	Adjusted p-value	Significant
Chautauqua - Erie	-5.743	0.0000	Yes
Dutchess - Erie	-6.118	0.0000	Yes
Dutchess - Fulton	-4.738	0.0000	Yes
Albany - Jefferson	5.740	0.0000	Yes
Erie - Jefferson	6.604	0.0000	Yes
Fulton - Jefferson	5.429	0.0000	Yes
Erie - Monroe	6.625	0.0000	Yes
Jefferson - Montgomery	-4.691	0.0000	Yes
Albany - Nassau	4.890	0.0000	Yes
Erie - Nassau	6.222	0.0000	Yes
Erie - Niagara	5.685	0.0000	Yes
Cayuga - Oneida	-4.583	0.0000	Yes
Chautauqua - Oneida	-4.606	0.0000	Yes
Dutchess - Oneida	-5.133	0.0000	Yes
Jefferson - Oneida	-5.778	0.0000	Yes
Nassau - Oneida	-4.929	0.0000	Yes
Cayuga - Onondaga	-4.566	0.0000	Yes
Chautauqua - Onondaga	-4.657	0.0000	Yes
Dutchess - Onondaga	-5.174	0.0000	Yes
Jefferson - Onondaga	-5.806	0.0000	Yes
Monroe - Onondaga	-4.626	0.0000	Yes
Nassau - Onondaga	-5.028	0.0000	Yes
Albany - Orange	4.519	0.0000	Yes
Erie - Orange	6.201	0.0000	Yes
Oneida - Orange	4.557	0.0000	Yes
Onondaga - Orange	4.716	0.0000	Yes
Erie - Rensselaer	5.535	0.0000	Yes
Albany - Rockland	7.873	0.0000	Yes
Erie - Rockland	9.175	0.0000	Yes
Fulton - Rockland	7.059	0.0000	Yes
Monroe - Rockland	5.273	0.0000	Yes
Montgomery - Rockland	6.102	0.0000	Yes
Oneida - Rockland	7.878	0.0000	Yes
Onondaga - Rockland	8.090	0.0000	Yes
Rockland - Schenectady	-4.675	0.0000	Yes
Rockland - Steuben	-4.764	0.0000	Yes
Albany - Westchester	10.520	0.0000	Yes
Erie - Westchester	13.938	0.0000	Yes
Fulton - Westchester	8.086	0.0000	Yes
Monroe - Westchester	6.646	0.0000	Yes
Montgomery - Westchester	6.664	0.0000	Yes
Niagara - Westchester	4.528	0.0000	Yes
Oneida - Westchester	10.364	0.0000	Yes
Onondaga - Westchester	11.498	0.0000	Yes
Orange - Westchester	4.618	0.0000	Yes
Schenectady - Westchester	4.966	0.0000	Yes
Steuben - Westchester	4.691	0.0000	Yes
Cayuga - Fulton	-4.357	0.0001	Yes
Chautauqua - Fulton	-4.184	0.0001	Yes
Albany - Monroe	4.306	0.0001	Yes

Table 3: Dunn post-hoc test: significant county pairs (BH-corrected)
(continued)

Comparison	Z statistic	Adjusted p-value	Significant
Fulton - Nassau	4.397	0.0001	Yes
Albany - Niagara	4.180	0.0001	Yes
Monroe - Oneida	-4.335	0.0001	Yes
Niagara - Oneida	-4.229	0.0001	Yes
Niagara - Onondaga	-4.324	0.0001	Yes
Albany - Rensselaer	4.194	0.0001	Yes
Oneida - Rensselaer	4.245	0.0001	Yes
Onondaga - Rensselaer	4.308	0.0001	Yes
Columbia - Rockland	4.316	0.0001	Yes
Niagara - Rockland	4.202	0.0001	Yes
Orange - Rockland	4.213	0.0001	Yes
Rockland - Warren	-4.414	0.0001	Yes
Broome - Westchester	4.382	0.0001	Yes
Columbia - Westchester	4.240	0.0001	Yes
Warren - Westchester	4.198	0.0001	Yes
Broome - Rockland	4.147	0.0002	Yes
Rockland - Ulster	-4.046	0.0002	Yes
Ulster - Westchester	4.062	0.0002	Yes
Broome - Oneida	-3.966	0.0003	Yes
Broome - Onondaga	-4.015	0.0003	Yes
Erie - Suffolk	3.977	0.0003	Yes
Albany - Broome	3.910	0.0004	Yes
Dutchess - Montgomery	-3.929	0.0004	Yes
Fulton - Orange	3.929	0.0004	Yes
Erie - Ulster	3.878	0.0004	Yes
Jefferson - Warren	-3.873	0.0004	Yes
Rensselaer - Westchester	3.914	0.0004	Yes
Rensselaer - Rockland	3.821	0.0005	Yes
Jefferson - Steuben	-3.823	0.0005	Yes
Fulton - Rensselaer	3.797	0.0006	Yes
Fulton - Niagara	3.717	0.0008	Yes
Erie - Schenectady	3.690	0.0009	Yes
Cayuga - Montgomery	-3.661	0.0010	Yes
Broome - Fulton	-3.550	0.0014	Yes
Fulton - Monroe	3.551	0.0014	Yes
Jefferson - Monroe	-3.556	0.0014	Yes
Fulton - Suffolk	3.556	0.0014	Yes
Oneida - Suffolk	3.563	0.0014	Yes
Albany - Suffolk	3.517	0.0015	Yes
Montgomery - Nassau	3.481	0.0017	Yes
Orleans - Rockland	3.481	0.0017	Yes
Onondaga - Suffolk	3.496	0.0017	Yes
Jefferson - Schenectady	-3.362	0.0026	Yes
Chautauqua - Montgomery	-3.320	0.0030	Yes
Dutchess - Warren	-3.327	0.0030	Yes
Columbia - Jefferson	3.305	0.0031	Yes
Cayuga - Warren	-3.290	0.0033	Yes
Erie - Oswego	3.265	0.0035	Yes
Orleans - Westchester	3.235	0.0039	Yes
Herkimer - Rockland	3.224	0.0040	Yes

Table 3: Dunn post-hoc test: significant county pairs (BH-corrected)
(continued)

Comparison	Z statistic	Adjusted p-value	Significant
Suffolk - Warren	-3.199	0.0043	Yes
Chemung - Rockland	3.165	0.0048	Yes
Otsego - Rockland	3.166	0.0048	Yes
Nassau - Rockland	3.099	0.0059	Yes
Montgomery - Suffolk	3.097	0.0059	Yes
Jefferson - Orleans	-3.071	0.0063	Yes
Dutchess - Steuben	-3.078	0.0063	Yes
Cortland - Erie	-3.045	0.0069	Yes
Chautauqua - Rockland	3.031	0.0071	Yes
Nassau - Warren	-2.981	0.0082	Yes
Oneida - Ulster	2.983	0.0083	Yes
Cayuga - Steuben	-2.969	0.0085	Yes
Otsego - Westchester	2.959	0.0087	Yes
Herkimer - Westchester	2.943	0.0091	Yes
Montgomery - Orange	2.936	0.0092	Yes
Chautauqua - Warren	-2.936	0.0092	Yes
Onondaga - Ulster	2.919	0.0095	Yes
Albany - Ulster	2.913	0.0096	Yes
Nassau - Westchester	2.919	0.0096	Yes
Fulton - Oswego	2.898	0.0100	Yes
Montgomery - Rensselaer	2.891	0.0101	Yes
Jefferson - Ulster	-2.887	0.0101	Yes
Jefferson - Otsego	-2.888	0.0102	Yes
Oneida - Oswego	2.870	0.0106	Yes
Chemung - Westchester	2.858	0.0109	Yes
Jefferson - Niagara	-2.842	0.0114	Yes
Fulton - Ulster	2.841	0.0114	Yes
Broome - Jefferson	2.833	0.0115	Yes
Rockland - Sullivan	-2.834	0.0115	Yes
Albany - Oswego	2.821	0.0118	Yes
Jefferson - Orange	-2.813	0.0120	Yes
Chautauqua - Westchester	2.800	0.0124	Yes
Cattaraugus - Rockland	2.795	0.0125	Yes
Onondaga - Oswego	2.791	0.0126	Yes
Montgomery - Niagara	2.770	0.0133	Yes
Oswego - Warren	-2.757	0.0138	Yes
Cortland - Fulton	-2.746	0.0141	Yes
Steuben - Suffolk	2.727	0.0148	Yes
Herkimer - Jefferson	2.728	0.0149	Yes
Cortland - Warren	-2.714	0.0153	Yes
Oneida - Schenectady	2.704	0.0156	Yes
Cortland - Oneida	-2.700	0.0157	Yes
Rensselaer - Warren	-2.679	0.0166	Yes
Albany - Cortland	2.654	0.0178	Yes
Orange - Warren	-2.649	0.0179	Yes
Broome - Montgomery	-2.645	0.0180	Yes
Onondaga - Schenectady	2.631	0.0187	Yes
Albany - Schenectady	2.627	0.0188	Yes
Cortland - Onondaga	-2.622	0.0189	Yes
Sullivan - Westchester	2.621	0.0189	Yes

Table 3: Dunn post-hoc test: significant county pairs (BH-corrected)
(continued)

Comparison	Z statistic	Adjusted p-value	Significant
Orleans - Suffolk	2.607	0.0195	Yes
Nassau - Steuben	-2.597	0.0200	Yes
Otsego - Suffolk	2.594	0.0200	Yes
Niagara - Warren	-2.592	0.0200	Yes
Cayuga - Orleans	-2.578	0.0205	Yes
Dutchess - Orleans	-2.580	0.0206	Yes
Jefferson - Rensselaer	-2.571	0.0208	Yes
Fulton - Schenectady	2.566	0.0210	Yes
Dutchess - Monroe	-2.559	0.0212	Yes
Jefferson - Sullivan	-2.560	0.0212	Yes
Broome - Warren	-2.553	0.0214	Yes
Cayuga - Otsego	-2.527	0.0230	Yes
Dutchess - Otsego	-2.516	0.0234	Yes
Rockland - Yates	-2.518	0.0234	Yes
Chautauqua - Steuben	-2.513	0.0235	Yes
Columbia - Dutchess	2.473	0.0259	Yes
Monroe - Montgomery	-2.474	0.0260	Yes
Montgomery - Oswego	2.453	0.0273	Yes
Cattaraugus - Westchester	2.449	0.0274	Yes
Dutchess - Schenectady	-2.430	0.0287	Yes
Chemung - Erie	-2.422	0.0290	Yes
Rockland - St. Lawrence	-2.424	0.0290	Yes
Chemung - Jefferson	2.392	0.0314	Yes
Monroe - Warren	-2.388	0.0315	Yes
Cayuga - Columbia	-2.383	0.0317	Yes
Cortland - Montgomery	-2.347	0.0349	Yes
Ulster - Warren	-2.336	0.0355	Yes
Cayuga - Monroe	-2.337	0.0356	Yes
Erie - Ontario	2.306	0.0380	Yes
Cortland - Otsego	-2.306	0.0382	Yes
Westchester - Yates	-2.299	0.0383	Yes
Oswego - Otsego	-2.300	0.0384	Yes
Cayuga - Schenectady	-2.291	0.0389	Yes
Suffolk - Sullivan	-2.283	0.0395	Yes
Nassau - Otsego	-2.278	0.0398	Yes
Nassau - Orleans	-2.264	0.0411	Yes
Chautauqua - Otsego	-2.260	0.0413	Yes
Jefferson - Yates	-2.249	0.0423	Yes
Chautauqua - Orleans	-2.235	0.0436	Yes
Ontario - Warren	-2.225	0.0443	Yes
Orleans - Oswego	2.222	0.0444	Yes
Columbia - Suffolk	2.226	0.0444	Yes
Cortland - Orleans	-2.215	0.0451	Yes
Cayuga - Sullivan	-2.199	0.0467	Yes
Dutchess - Sullivan	-2.185	0.0479	Yes
Herkimer - Suffolk	2.186	0.0480	Yes
Cattaraugus - Erie	-2.168	0.0497	Yes

```
# calculating effect size  
kw_result$statistic/(nrow(non0rate_data)-1)
```

Kruskal-Wallis chi-squared 0.05576575

5 Conclusion

While this data does demonstrate some interesting results that show disparities across New York State counties, we can not say with certainty that these results are solely due to difference between county. Further investigation should hinge on better data collection tools to capture all children within New York State; this only relies on children who are able to go to a medical clinic and perform blood lead level testing. Individuals who are under- or not insured, undocumented, or unable to go to the clinic themselves are not captured within this data.

Furthermore, this data presents a narrative about disparities within pediatric lead, however additional sociodemographic data is needed to better contextualize why some counties might have higher/lower rates. Adding in when specific New York State legislation and major waterworks infrastructural projects were initialized can provide additional context to if and when lead levels are related to environmental, infrastructural, and/or other sources of contamination.

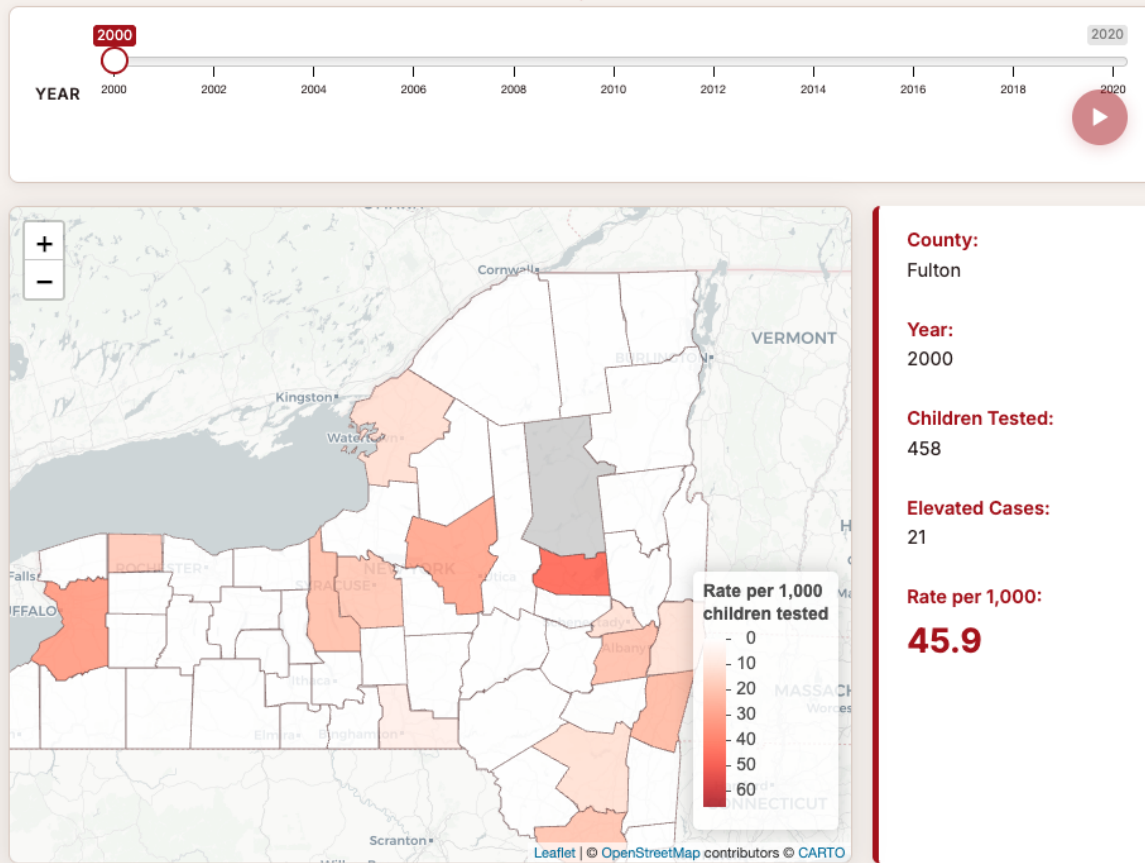
6 Shiny App

This data can be further explored by downloading information within [this GitHub repository](#). For an interactive way to see between-county differences, [this ShinyApp](#) was created for anyone to view and access. Instructions on how to recreate, edit, or access this data using this ShinyApp is also available on GitHub.

Childhood Blood Lead Levels — New York State (excluding NYC)

Elevated blood lead incidence rate per 1,000 children tested, by county

Scroll down for background, data sources, and disclosures



7 References

1. CDC. CDC Updates Blood Lead Reference Value. Childhood Lead Poisoning Prevention. Published April 2, 2024. <https://www.cdc.gov/lead-prevention/php/news-features/updates-blood-lead-reference-value.html>
2. New York State Department of Health. Childhood Lead Poisoning Prevention. ny.gov. Accessed February 19, 2026. <https://www.health.ny.gov/environmental/lead/>
3. Lead Service Lines in NYC Disproportionately Impact Hispanic/Latino Communities and Children Already At Risk. Columbia University Mailman School of Public Health. Published August 30, 2023. <https://www.publichealth.columbia.edu/news/lead-service-lines-nyc-disproportionately-impact-hispanic-latino-communities-children-already-risk-lead-exposure>

8 AI disclosure statement

The author used Claude (Anthropic, claude.ai) as an AI assistant during the preparation of this analysis. AI assistance was used for the following purposes: debugging R code and checking interpretations for statistical

output. All statistical analyses were performed in R, and all text was written by the author without gAI input.